

Exhibit 5.67 provides capital market assumptions for five asset classes: US stocks, US REITs, the MSCI EAFE for international stocks, US bonds, and US bills. These include the arithmetic real returns, compounded returns, volatilities, and correlations. These are not my recommended assumptions—they are a case study about the process. Assuming that these values align with a retiree’s viewpoints and choices for asset allocation, Exhibit 5.26 provides the ingredients to study sustainable spending rates.

Exhibit 5.26

Hypothetical Long-Term Capital Market Expectations (in Real US Dollar Terms)

	Arithmetic Means	Geometric Means	Standard Deviations	Correlation Coefficients				
				US Stocks	US REITS	MSCI EAFE	US Bonds	US Bills
US Stocks	6.5%	4.50%	20%	1	0.7	0.6	0.1	0.1
US REITs	5.5%	4.22%	16%	0.7	1	0.5	0.3	0.4
MSCI EAFE	6.5%	4.08%	22%	0.6	0.5	1	−0.1	−0.1
US Bonds	0.7%	0.46%	7%	0.1	0.3	−0.1	1	0.7
US Bills	−0.3%	−0.38%	4%	0.1	0.4	0	0.7	1

Note: These are hypothetical numbers used to illustrate the framework for expanding the asset allocation that can be used with sustainable spending studies. They do not represent actual forecasts.

To assess the importance of these different capital market expectations, I compute an efficient frontier for this set of assumptions. Software is available to make such computations, as this task is generally not done by hand. Mean-variance optimizer software uses inputs to find the efficient frontier of asset allocations that provide the most return for a given volatility or the least volatility for a given return.

The efficient frontier can then be plotted onto the grid framework relating withdrawal rates to returns and volatilities for an acceptable failure probability and retirement duration. In Exhibit 5.27, I overlaid the efficient frontier (a lighter color for only considering US stocks and bonds, black for all five asset classes in Exhibit 5.26) onto the withdrawal rate charts for a thirty-year retirement duration and a 10 percent acceptable failure probability (different